

TITANIC SURVIVAL PREDICTION USING MACHINE LEARNING

1. ABSTRACT

This project focuses on predicting passenger survival on the Titanic using Machine Learning techniques. The dataset includes passenger information such as age, gender, ticket class, fare, and embarkation point. The objective is to build a classification model that accurately predicts whether a passenger survived or not.

2. INTRODUCTION

The Titanic disaster is one of the most famous shipwrecks in history. Machine Learning can be applied to analyze passenger data and identify patterns that influenced survival. This project demonstrates how classification algorithms can be used in real-world datasets.

3. PROBLEM STATEMENT

The goal is to develop a predictive model that determines whether a passenger survived the Titanic disaster based on available features. The problem is a binary classification task (Survived = 1, Not Survived = 0).

4. DATASET DESCRIPTION

The dataset is obtained from Kaggle Titanic dataset and includes the following features:

- PassengerId
- Survived (Target Variable)
- Pclass (Ticket class)

- Name
 - Sex
 - Age
 - SibSp (Siblings/Spouses aboard)
 - Parch (Parents/Children aboard)
 - Ticket
 - Fare
 - Cabin
 - Embarked
-

5. DATA PREPROCESSING

Data preprocessing steps include:

- Handling missing values in Age, Cabin, and Embarked
 - Encoding categorical variables such as Sex and Embarked
 - Removing irrelevant columns like Name, Ticket, and Cabin
 - Feature transformation and scaling
-

6. EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed to understand relationships between variables:

- Survival rate comparison based on gender
- Survival rate based on passenger class
- Age distribution of passengers
- Correlation analysis between features
- Visualization using bar charts, histograms, and heatmaps

7. FEATURE ENGINEERING

New features were created to improve model performance:

- $\text{FamilySize} = \text{SibSp} + \text{Parch} + 1$
- Conversion of categorical variables into numerical format
- Selection of important features for training

8. MODEL BUILDING

Machine Learning algorithms applied:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Dataset is split into training and testing sets to evaluate model performance.

9. MODEL EVALUATION

The models were evaluated using:

- Accuracy Score
- Confusion Matrix
- Precision, Recall, and F1 Score

Among all models, Random Forest performed better in terms of accuracy and stability.

10. RESULTS AND DISCUSSION

The model successfully predicts survival with good accuracy. Key factors influencing survival are:

- Gender (Females had higher survival rate)
 - Passenger Class (Higher class had better survival chances)
 - Age (Children had higher survival probability)
-

11. CONCLUSION

This project demonstrates the application of Machine Learning in predicting survival outcomes. The model helps understand important factors affecting survival and achieves good prediction performance.

12. FUTURE ENHANCEMENTS

- Use advanced algorithms like XGBoost or Neural Networks
 - Perform hyperparameter tuning
 - Deploy model using Flask or Streamlit
 - Improve feature engineering techniques
-

13. REFERENCES

- Kaggle Titanic Dataset
- Scikit-learn Documentation
- Python Data Science Libraries